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Pyros: a raster–vector spatial simulation model for predicting wildland surface fire spread and growth

Debora Voltolina | Giacomo Cappellini | Tiziana Apuani | Simone Sterlacchini

Environmental Science · International journal of wildland fire · 8 March 2024

Background Euro–Mediterranean regions are expected to undergo a climate-induced exacerbation of **fire** activity in the upcoming decades. Reliable **predictions** of **fire** behaviour represent an essential… Expand

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Designing Operationally Relevant Daily Large Fire Containment Strategies Using Risk Assessment Results

Yu Wei | Matthew P. Thompson | Joe H. Scott | Christopher D. O’Connor | Christopher J. C. Dunn

Environmental Science, Engineering · Forests · 5 April 2019

TLDR This research expands the set of data inputs to include raster layers of simulated burn probability, flame length probability, **fire** arrival time, **and** expected net value change, all calculated using a common set of stochastic weather **forecasts and landscape** data **and** develops a secondary optimization **model**. Expand

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A High Resolution Coupled Fire–Atmosphere Forecasting System to Minimize the Impacts of Wildland Fires: Applications to the Chimney Tops II Wildland Event

P. Jiménez | D. Muñoz-Esparza | B. Kosović

Environmental Science, Engineering · 12 April 2018

Wildland fires are responsible for large socio-economic impacts. **Fires** affect the environment, damage structures, threaten lives, cause health issues, and involve large suppression costs. These… Expand

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Predicted changes in fire weather suggest increases in lightning fire initiation and future area burned in the mixedwood boreal forest

M. Krawchuk S. Cumming M. Flannigan Environmental Science · 2009

Forecasting future **fire** activity as a function of climate change is a step towards understanding the future state of the western mixedwood boreal ecosystem. We developed five annual weather indices… Expand

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Development of a Smoke Dispersion Forecast System for Korean Forest Fires

Boknam Lee S. Cho Seung-Kii Lee C. Woo Joowon Park

Environmental Science · Forests · 1 March 2019

Smoke from **forest fires** is a growing concern in Korea as **forest** structures have changed and become more vulnerable to **fires** associated with climate change. In this study, we developed a Korean **forest**… Expand

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Sensitivity of atypical lateral fire spread to wind and slope

C. Simpson J. Sharples J. Evans Environmental Science · 28 February 2016

This study presents new knowledge of the environmental sensitivity of a dynamic mode of atypical **wildland fire spread** on steep lee-facing slopes. This is achieved through a series of idealized… Expand

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Using simulation to map fire regimes: an evaluation of approaches, strategies, and limitations

R. Keane G. Cary R. Parsons Environmental Science · 24 December 2003

Spatial depictions of **fire** regimes are indispensable to **fire management** because they portray important characteristics of **wildland fire**, such as severity, intensity, and pattern, across a landscape… Expand

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Gradient modeling: A new approach to fire modeling and wilderness resource management

S. R. Kessell | Environmental Science · 1977

TLDR Gradient **model**ing, a new approach to resource **management and forest fire simulation**, has been developed to meet the needs of managers of **wild**erness resources in Glacier National Park. Expand

60 | Springer Nature | Save | Cite

A two-dimensional model of fire spread across a fuel bed including wind combined with slope conditions

F. Morandini | P. Santoni | J. Balbi | J. Ventura | J. Mendes-Lopes

Environmental Science, Engineering · 15 April 2002

In a previous work (Santoni et al., Int. J. **Wildland Fire**, 2000, 9(4), 285–292), we proposed a twodimensional **fire spread model** including slope effects as another step towards our aim to elaborate a… Expand

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